

Knowledge and Data

Working Group

Week 4: Expressive ontologies in OWL



Question 1—RDFS Shortcomings

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- Identity of instances (owl:sameAs)

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- Properties of properties (e.g. transitivity)
- Disjointness and equivalence of classes
- Distinctness of objects (`owl:differentFrom`)
- Identity of instances (`owl:sameAs`)
- Local restrictions
- Negative property assertions
- etc.

Question 2—OWL

Task: Name which of the following assumptions are characteristics of OWL:

- Unique world assumption
- Unique name assumption
- Open world assumption
- Open name assumption
- Multiple graph assumption
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Open World Assumption (OWA):

Everything that is not explicitly stated is assumed unknown (neither true nor false)

Question 3—OWL Classes

Task: Name which of the following classes is a valid OWL class:

- owl:Nothing
- owl:Half
- owl:Object
- owl:Thing
- owl:Resource
- owl:TransitiveThing

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- **owl:Thing**
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owl:Nothing is a subset of all classes, and represents the empty set ($\emptyset$).

owl:Thing is the most general class in OWL, and represents the “universe”. Every class is a subset of owl:Thing, and every individual is a member of this class.

Question 4—Modelling in OWL: equivalency and disjointness

Consider the following OWL ontology:

ex:C ₁	owl:equivalentClass	ex:C ₂ .
ex:C ₂	owl:disjointWith	ex:C ₃ .
ex:a ₁	rdf:type	ex:C ₁ .
ex:a ₂	rdf:type	ex:C ₂ .
ex:a ₃	rdf:type	ex:C ₃ .

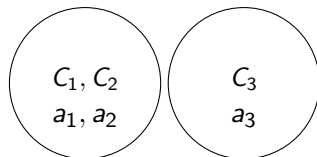
Which triples can be derived using OWL semantics?

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ex:a ₂	rdf:type	ex:C ₂ .
ex:a ₃	rdf:type	ex:C ₃ .

As a set:



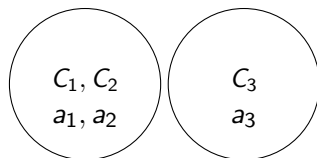
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ex:a ₂	rdf:type	ex:C ₂ .
ex:a ₃	rdf:type	ex:C ₃ .

As a set:



Which triples can be derived using OWL semantics?

Answer (partial):

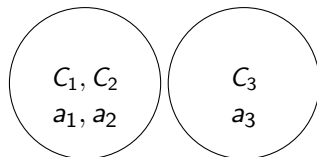
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ex:a ₂	rdf:type	ex:C ₂ .
ex:a ₃	rdf:type	ex:C ₃ .

As a set:



Which triples can be derived using OWL semantics?

Answer:

ex:a ₁	rdf:type	ex:C ₂ .
ex:a ₂	rdf:type	ex:C ₁ .
ex:a ₁	owl:differentFrom	ex:a ₃ .
ex:a ₂	owl:differentFrom	ex:a ₃ .

Question 5—Modelling in OWL: classes

Consider the following OWL ontology:

$\text{ex:C}_1 \quad \text{owl:oneOf} \quad (\text{ex:a}_1 \text{ ex:a}_3) .$
 $\text{ex:C}_2 \quad \text{owl:oneOf} \quad (\text{ex:a}_1 \text{ ex:a}_2) .$
 $\text{ex:C}_3 \quad \text{owl:intersection} \quad (\text{ex:C}_1 \text{ ex:C}_2) .$

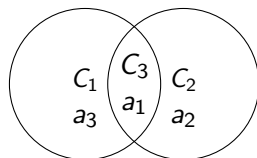
Which triples can be derived using OWL semantics?

Question 5—Modelling in OWL: classes

Consider the following OWL ontology:

ex:C₁ owl:oneOf (ex:a₁ ex:a₃) .
ex:C₂ owl:oneOf (ex:a₁ ex:a₂) .
ex:C₃ owl:intersection (ex:C₁ ex:C₂) .

As a set:



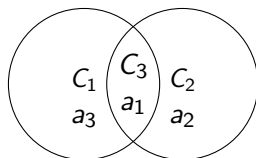
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ex:C₃ owl:intersection (ex:C₁ ex:C₂) .

As a set:



Which triples can be derived using OWL semantics?

Answer:

ex:a₁ rdf:type ex:C₁ ,
ex:C₂ ,
ex:C₃ .
ex:a₂ rdf:type ex:C₂ .
ex:a₃ rdf:type ex:C₁ .

Question 6—Modelling in OWL: properties

Task: Name which of the following properties is both *asymmetric* and *transitive*:

- fatherOf
- teacherOf
- siblingOf
- uncleOf
- ancestorOf

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asymmetric but not transitive

asymmetric (?) but not transitive

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Task: Name which of the following properties is both *asymmetric* and *transitive*:

- fatherOf asymmetric but not transitive
- teacherOf asymmetric (?) but not transitive
- siblingOf transitive but not asymmetric
- unclEoF
- ancestorOf

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| ■ siblingOf | transitive but not asymmetric |
| ■ unclEoF | asymmetric but not transitive |
| ■ ancestorOf | |

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| ■ teacherOf | asymmetric (?) but not transitive |
| ■ siblingOf | transitive but not asymmetric |
| ■ unclEoF | asymmetric but not transitive |
| ■ ancestorOf | asymmetric and transitive |

Only **ancestorOf** is both asymmetric and transitive

Question 7—Modelling in OWL: properties II

Consider the following OWL ontology:

ex:p	rdf:type	owl:FunctionalProperty .
ex:q	rdf:type	owl:InverseFunctionalProperty .
ex:a ₁	ex:p	ex:a ₂ ,
		ex:a ₃ .
ex:a ₃	ex:q	ex:a ₂ .

Which triples can be derived using OWL semantics?

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ex:a ₁	ex:p	ex:a ₂ ,
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ex:a ₃	ex:q	ex:a ₂ .

Which triples can be derived using OWL semantics?

Answer (partial):

ex:a ₂	owl:sameAs	ex:a ₃ .
ex:a ₃	owl:sameAs	ex:a ₂ ;

Question 7—Modelling in OWL: properties II

Consider the following OWL ontology:

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ex:p    rdf:type    owl:FunctionalProperty .
ex:q    rdf:type    owl:InverseFunctionalProperty .
ex:a1  ex:p        ex:a2 ,
                        ex:a3 .
ex:a3  ex:q        ex:a2 .
```

Which triples can be derived using OWL semantics?

Answer:

```
ex:a2  owl:sameAs  ex:a3 .
ex:a3  owl:sameAs  ex:a2 ;
                        ex:q        ex:a3 .
ex:a2  ex:q        ex:a2 ,
                        ex:a3 .
```

Question 8—Modelling in OWL: Property Axioms

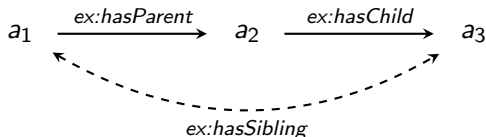
Task: Explain the concept of the *property chain* and how you can use it to model the *hasSibling* relation.

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OWL's property chain:

An expression that composes two or more properties

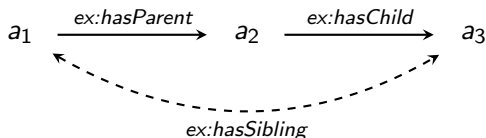


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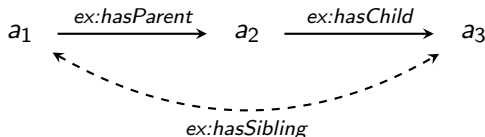
```
ex:hasSibling    a                owl:ObjectProperty ;  
                owl:propertyChainAxiom (ex:hasParent ex:hasChild) ;
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OWL's property chain:

An expression that composes two or more properties



ex:hasSibling	a	owl:ObjectProperty ;
	owl:propertyChainAxiom	(ex:hasParent ex:hasChild) ;
	a	owl:IrreflexiveProperty ,
		owl:SymmetricProperty .

Question 9—OWL Inferencing

Consider the following OWL ontology:

ex:a owl:sameAs ex:b .
ex:b owl:sameAs ex:c .

And **only** the following OWL entailment rule:

$$v \text{ owl:sameAs } w \rightarrow w \text{ owl:sameAs } v$$

Which triples can be derived using OWL semantics?

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And **only** the following OWL entailment rule:

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Which triples can be derived using OWL semantics?

Answer:

```
ex:b    owl:sameAs    ex:a .  
ex:c    owl:sameAs    ex:b .
```

Question 10—OWL Inferencing II

Consider the following OWL ontology:

ex:Footballer	rdfs:subClassOf	ex:Sportsman ;	
	owl:equivalentClass	[rdf:type owl:Restriction ;	
		owl:onProperty	ex:playsMatch ;
		owl:someValuesFrom	ex:FootballMatch .
		]	.
ex:john	ex:playsMatch	ex:match123 .	
ex:match123	rdf:type	ex:FootballMatch .	

Which triples can be derived using OWL semantics?

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Consider the following OWL ontology:

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ex:john	ex:playsMatch	ex:match123 .	
ex:match123	rdf:type	ex:FootballMatch .	

Which triples can be derived using OWL semantics?

Necessary and Sufficient: Class Footballer has exactly the same members
as the anonymous class

NB: equivalence not the same as class equality!

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ex:john	ex:playsMatch	ex:match123 .	
ex:match123	rdf:type	ex:FootballMatch .	

Which triples can be derived using OWL semantics?

Any statement $s \text{ p } o$ in which p equals `ex:playsMatch` and o is of type `ex:FootballMatch` results in s being inferred as a member of the anonymous class.

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ex:john	ex:playsMatch	ex:match123 .	
ex:match123	rdf:type	ex:FootballMatch .	
ex:john	rdf:type	ex:Footballer , ex:Sportsman .	

Which triples can be derived using OWL semantics?

Any statement $s \text{ p } o$ in which p equals `ex:playsMatch` and o is of type `ex:FootballMatch` results in s being inferred as a member of the anonymous class.

Any member of the anonymous class is also a member of the class `ex:Footballer` (since these classes are equivalent) and any member of the `ex:Footballer` class is also a member of the `ex:Sportsman` class (because the latter is a superclass of the former).

Question 11—OWL Inferencing II

Consider the following OWL ontology:

ex:A	rdf:type	owl:Class ;	
	owl:equivalentClass	[rdf:type owl:Restriction ;	
		owl:onProperty	ex:p ;
		owl:allValuesFrom	ex:B .
		]	.
ex:p	rdf:type	owl:ObjectProperty .	
ex:b	rdf:type	ex:B .	
ex:a	ex:p	ex:b .	

Which triples, if any, can be inferred using OWL semantics?

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Which triples, if any, can be inferred using OWL semantics?

For any individual s to be a member of the anonymous class, it must hold that, for each statement $s \text{ p } o$ in which p equals ex:p , o must be of type B .

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ex:a	ex:p	ex:b .	

Which triples, if any, can be inferred using OWL semantics?

For any individual s to be a member of the anonymous class, it must hold that, for each statement $s \text{ p } o$ in which p equals ex:p , o must be of type B .

Answer:

nothing can be inferred

Question 11—OWL Inferencing II

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Why is it not possible to infer that `ex:a rdf:type ex:A .` ?

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ex:b	rdf:type	ex:B .	
ex:a	ex:p	ex:b .	

Why is it not possible to infer that `ex:a rdf:type ex:A .` ?

Answer:

To infer that `ex:a rdf:type ex:A`, it is necessary for `ex:a` to be a member of the anonymous class, but this requires us to know that, for all statements `ex:a ex:p o`, the object `o` is of type `ex:B`. This requires a closed view of the world, whereas OWL maintains an open view (OWA)! Likewise, it is also impossible to infer that `ex:a` is *not* a member of class `ex:A`.

Question 12—Modelling in OWL

Consider the following OWL ontology:

ex:A	rdfs:subClassOf	ex:B .
ex:B	owl:disjointWith	ex:C .
ex:p	rdfs:domain	ex:A ;
	rdfs:range	ex:B .
ex:q	rdfs:subPropertyOf	ex:p .
ex:a	rdf:type	ex:A .
ex:b	rdf:type	ex:B .
ex:c	rdf:type	ex:C .
ex:b	ex:q	ex:a ,
		ex:c .

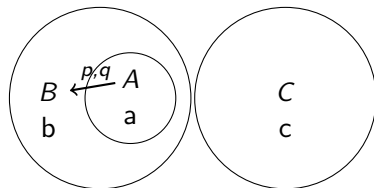
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ex:q	rdfs:subPropertyOf	ex:p .
ex:a	rdf:type	ex:A .
ex:b	rdf:type	ex:B .
ex:c	rdf:type	ex:C .
ex:b	ex:q	ex:a ,
		ex:c .

As a set:



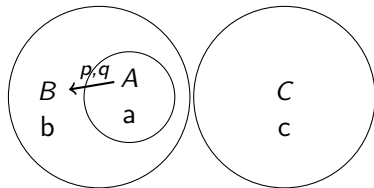
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ex:p	rdfs:domain	ex:A ;
	rdfs:range	ex:B .
ex:q	rdfs:subPropertyOf	ex:p .
ex:a	rdf:type	ex:A .
ex:b	rdf:type	ex:B .
ex:c	rdf:type	ex:C .
ex:b	ex:q	ex:a ,
		ex:c .

As a set:



Which triples can be derived using OWL semantics?

Answer: None, as the ontology is inconsistent:

The range of `ex:q` is an (anonymous) subclass of the range of its superproperty: $ex:_D \subseteq ex:B$. Because also `ex:b` `ex:q` `ex:c`, this implies that `ex:c` is a member of this subclass. But since `ex:c` is already of type `ex:C` and since classes `ex:B` (and its subclasses) and `ex:C` are disjoint, this cannot be.

Quiz

To test your proficiency:

Exercise quiz Module 4
(see the course's Canvas page)

NB: completing the quiz is mandatory to continue to next week's module